ROHIT SEN

LEAD SOFTWARE ENGINEER | FULL-STACK & AI/ML ENGINEER

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OVERVIEW

Lead Software Engineer with experience spanning full-stack application development, backend engineering, software architecture and AI/ML-powered solutions. Skilled in building web applications, APIs, data-driven platforms and intelligent automation workflows using modern development technologies.

AI/ML Engineer with hands-on experience building data-driven and AI-powered applications, spanning machine learning, deep learning, NLP, generative AI, and full-stack development. Skilled in designing intelligent systems, automating workflows, and delivering scalable, production-ready solutions using Python, React, and modern cloud/deployment tools.

SKILLS

Data Analysis: Proficient in SQL, Power BI, MS Excel, and Python for data analysis and reporting.
Data Visualization: Experienced in creating dashboards and visual reports using Power BI.
Database Management: Good knowledge of SQL for database design and management.
Programming: Knowledge of Python.
Machine Learning: Knowledge of machine learning algorithms and techniques for building models using Python libraries like scikit-learn, etc.
Data Manipulation: Good knowledge of data manipulation using Python libraries like NumPy, Pandas, etc.
Data Wrangling: Knowledge of the data wrangling process for converting raw data into structured and usable data.
Generative AI: Knowledge of Large Language Models (LLMs) and their applications in text generation, summarization, question answering, and chatbot development.
Agentic AI: Understanding of building autonomous AI agents capable of task planning, reasoning, and tool usage.
Retrieval-Augmented Generation (RAG): Knowledge of implementing RAG pipelines by integrating LLMs with vector databases for context-aware and knowledge-grounded responses.
LangChain: Knowledge of LangChain for building AI-powered applications.
LLM Integration: Experience working with OpenAI APIs and Python frameworks for building AI-powered applications.
FastAPI: Knowledge of building APIs, handling requests and responses, and integrating backend services for AI/ML applications.
Deployment & Version Control: Hands-on knowledge of deploying applications on Render, Vercel, Netlify, and Railway, along with Git and GitHub for version control.
AI Tools: Knowledge of AI-assisted development tools such as Codex, Antigravity, and Claude Code.
MongoDB: Knowledge of MongoDB and Mongoose for NoSQL database design, schema modeling, and data management.
Cloudinary: Experience integrating Cloudinary for media/image upload, storage, and management in web applications.
Docker: Knowledge of containerizing applications using Docker for consistent development and deployment environments.
AWS: Basic knowledge of deploying and hosting applications on AWS (EC2, S3) for scalable cloud infrastructure.

Additional Software Engineering Skills

Frontend: React.js, Next.js, TypeScript, JavaScript, HTML5, CSS3, Tailwind CSS, Flutter
Backend & APIs: Node.js, Express.js, NestJS, Flask, Django, REST APIs, JWT, API integration
Databases: MySQL, PostgreSQL, MongoDB Atlas, DynamoDB, Amazon RDS, Redis, Elasticsearch
AI/ML Stack: TensorFlow, PyTorch, Hugging Face, Google Gemini API, Sentence Transformers, NLP, Computer Vision, embeddings, semantic search, ChromaDB, FAISS
Cloud & DevOps: Azure Data Factory, ADLS Gen2, Kubernetes, Jenkins, Terraform, Ansible, CI/CD
Analytics & Visualization: Tableau, Amazon QuickSight, Streamlit, EDA, data reporting
3D & Motion: Three.js, React Three Fiber, GSAP, Framer Motion
Engineering Tools: Git, GitHub, Render, Vercel, Netlify, Railway, Cloudinary, Codex, Antigravity, Claude Code

WORK EXPERIENCE

Lead Software Developer

Oct 2026 – Present

Black Grapes Softech | Indore, Madhya Pradesh, India | Full-time | On-site

- Leading end-to-end software development, system architecture and technical execution of scalable, high-performance applications.
- Designing, developing and deploying full-stack applications, backend services and REST APIs using Python, Node.js, React.js, Next.js and FastAPI.
- Working with SQL, PostgreSQL and MongoDB databases, AI/ML integrations, cloud infrastructure and DevOps practices.
- Driving software quality, security, technical problem-solving, code reviews and collaboration with clients and stakeholders.
- Providing technical guidance to developers and supporting architecture and implementation decisions.

Software Engineer
Aug 2026 – Present

04 O'Clock Technologies | Remote | Freelance

Professional Freelancer
Apr 2026 – Present

Self-employed | Remote | Freelance

AI/ML Engineer
May 2026 – Sep 2026

Zenvora Pvt. Ltd.

- Developed and integrated AI/ML solutions using Python, NLP, LLMs, and semantic analysis to automate complex business workflows and improve decision-making.
- Built LLM-powered applications and RAG pipelines for context-aware question answering, combining document retrieval, contextual information, and generative AI responses.
- Developed intelligent document and text-processing pipelines for information extraction, entity/skill detection, classification, similarity analysis, and structured data generation.
- Implemented semantic matching and ranking systems using embeddings, similarity techniques, and NLP-based feature extraction to evaluate relevance between unstructured documents and predefined requirements.
- Designed AI-driven automation workflows that combine ML/LLM inference with backend APIs, business logic, and downstream actions, reducing manual processing.
- Developed and integrated FastAPI-based AI services with databases and frontend applications, exposing ML/LLM capabilities through scalable REST APIs.
- Worked with Google Gemini, Sentence Transformers, Scikit-learn, NLP techniques, MongoDB, and Python-based AI pipelines to build production-oriented AI features.
- Integrated AI solutions with external services and automated communication workflows, including email, calendar, and messaging APIs.
- Focused on building production-ready AI applications, including prompt engineering, model integration, response handling, error handling, API integration, and deployment-oriented architecture.

Data Analysis Intern
Feb 2025 – May 2025

Unified Mentor Private Limited

- Analysed and cleaned large datasets using Excel and Python, improving data accuracy and efficiency by 25%.
- Created interactive Power BI dashboards and visual reports, leading to improved decision-making for business stakeholders.
- Wrote complex SQL queries to extract insights from relational databases, reducing manual reporting time by 40%.

PROJECTS

AI-Powered HRMS & Recruitment Automation System

Tech Stack: *React.js, TypeScript, Python, FastAPI, MongoDB, Gemini AI, Scikit-learn*

- Developed an enterprise HRMS platform covering employee management, attendance, leave, recruitment, onboarding, performance, productivity, interviews, and document management using React.js, TypeScript, FastAPI, Python, and MongoDB.
- Built an AI-powered resume screening and candidate ranking pipeline for PDF/DOCX parsing, skill & experience extraction, JD matching, semantic similarity analysis, job-fit scoring, risk analysis, and automated candidate shortlisting.
- Implemented a recruiter dashboard displaying candidate match scores, matched/missing skills, experience, AI recommendations, and Shortlisted/Review/Rejected workflows, reducing manual resume screening effort.
- Integrated Google Gemini, WhatsApp, email, and Google Calendar for AI-assisted HR interactions, candidate communication, interview scheduling, shortlist reporting, and automated recruitment workflows.

Weather Impact on Crop Yield Prediction

- Developed a machine learning-based web application using Python and Streamlit to predict crop yields based on weather parameters (temperature, humidity, rainfall), achieving an R^2 score of 0.92.
- Processed and visualized 10,000+ data points across 28 Indian states to identify key correlations between climatic conditions and agricultural productivity.
- Enabled real-time crop yield prediction with 85%+ accuracy, empowering stakeholders with actionable insights for climate-resilient farming decisions.

Lead Scraper System

- Built a Python-based application to collect and process lead data from multiple online platforms using APIs.
- Stored processed lead information in MongoDB, enabling efficient querying and historical data tracking.
- Automated lead extraction workflows to reduce manual data collection and improve efficiency.
- Enabled export of cleaned lead data for further analysis and reporting.

TechMaster — Enterprise CMS & 3D Web Site

Tech Stack: *React, TypeScript, Node.js, Express.js, MongoDB, Mongoose, Three.js, React Three Fiber, GSAP, JWT, Cloudinary*

- Developed a full-stack enterprise CMS platform with React/TypeScript frontend, Node.js/Express REST APIs, MongoDB, JWT authentication, and role-based access control for centralized website content management.
- Built 40+ CMS modules and REST API workflows covering services, portfolio, blogs, careers, media, FAQs, testimonials, SEO, users, analytics, and website settings, enabling dynamic content delivery to the public website.
- Engineered API-driven data synchronization and media workflows using MongoDB, Mongoose, Cloudinary, and Multer, replacing hardcoded content with dynamically managed database-backed content.
- Developed a responsive 3D interactive web experience using Three.js, React Three Fiber, GSAP, Framer Motion, and Tailwind CSS, implementing animated 3D scenes, scroll interactions, transitions, and reusable components across 20+ pages.

PROFESSIONAL ACHIEVEMENTS

- Earned certification in "Data Science for Engineers" from NPTEL (IIT), demonstrating strong foundational and practical understanding of data science concepts.
- Completed industry-recognized certifications from Accenture and Tata, applying real-world data analytics and visualization techniques.

EDUCATION

Bachelor of Technology in Computer Science Engineering – Data Science

RGPV University, Bhopal | 2026

- Relevant Courses: Data Analysis, Database Management, Python Programming, Machine Learning.
- Certifications: Certified in Data Analytics by Accenture, Certified in Data Visualization by Tata, and Data Science for Engineers Certification by NPTEL.